

$$\hat{\mathbf{Q}}_{t,k}^1 = \mathbf{Q}_t^{\tau_k^Q} + (1 - \tau_k^Q) v_{\theta_Q}^Q \left(\mathbf{Q}_t^{\tau_k^Q}, \tau_k^Q; \mathcal{K}_{t,k}^Q \right),$$

$$\mathbf{e}_{t,k}^Q = \mathbf{P}^Q \mathbf{W}_{t,k}^Q \left(\mathbf{Y}_{t,k}^Q - \hat{\mathbf{Q}}_{t,k}^1 \right),$$

$$v_{\text{FBFM}}^Q = v_{\theta_Q}^Q + \lambda_{\tau_k^Q}^Q \left(\mathbf{J}_{t,k}^Q \right)^\top \mathbf{e}_{t,k}^Q$$